

BLADES PLC CASE STUDY

Assessment of potential arbitrage opportunities

Recall that Blades, a UK manufacturer of roller blades, has chosen Thailand as its primary export target for 'Speedos', Blades' primary product. Moreover, Blades' primary customer in Thailand, Entertainment Products, has committed itself to purchase 180 000 Speedos annually for the next three years at a fixed price denominated in baht, Thailand's currency. Because of quality and cost considerations, Blades also imports some of the rubber and plastic components needed to manufacture Speedos.

Lately, Thailand has experienced weak economic growth and political uncertainty. As investors lost confidence in the Thai baht as a result of the political uncertainty, they withdrew their funds from the country. This resulted in an excess supply of baht for sale over the demand for baht in the foreign exchange market, which put downward pressure on the baht's value. As foreign investors continued to withdraw their funds from Thailand, the baht's value continued to deteriorate. Since Blades has net cash flows in baht resulting from its exports to Thailand, a deterioration in the baht's value will affect the company negatively.

Ben Holt, Blades' finance director, would like to ensure that the spot and forward rates Blades' bank has quoted are reasonable. If the exchange rate quotes are reasonable, then arbitrage will not be possible. If the quotations are not appropriate, however, arbitrage may be possible. Under these conditions, Holt would like Blades to use some form of arbitrage to take advantage of possible mispricing in the foreign exchange market. Although Blades is not an arbitrageur, Holt believes that arbitrage opportunities could offset the negative impact resulting from the baht's depreciation, which would otherwise seriously affect Blades' profit margins.

Ben Holt has identified three arbitrage opportunities as profitable and would like to know which one of them is the most profitable. Thus, he has asked you, Blades' financial analyst, to prepare an analysis of the arbitrage opportunities he has identified. This would allow Holt to assess the profitability of arbitrage opportunities very quickly.

- 1 The first arbitrage opportunity relates to locational arbitrage. Holt has obtained spot rate quotations

from two banks in Thailand: Minzu Bank and Sobat Bank, both located in Bangkok. The bid and ask prices of Thai baht for each bank are displayed in the table below:

	Minzu Bank	Sobat Bank
Bid	£0.0149	£0.0152
Ask	£0.0151	£0.0153

Determine whether the foreign exchange quotations are appropriate. If they are not appropriate, determine the profit you could generate by withdrawing £100 000 from Blades' cheque account and engaging in arbitrage before the rates are adjusted.

- 2 Besides the bid and ask quotes for the Thai baht provided in the previous question, Minzu Bank has provided the following quotations for the pound and the Japanese yen:

	Quoted Bid Price	Quoted Ask Price
Value of a Japanese yen in British pounds	£0.0057	£0.0058
Value of a Thai baht in Japanese yen	¥2.69	¥2.70

Determine whether the cross exchange rate between the Thai baht and Japanese yen is appropriate. If it is not appropriate, determine the profit you could generate for Blades by withdrawing £100 000 from Blades' current account and engaging in triangular arbitrage before the rates are adjusted.

- 3 Ben Holt has obtained several forward contract quotations for the Thai baht to determine whether covered interest arbitrage may be possible. He was quoted a forward rate of £0.015 per Thai baht for a 90-day forward contract. The current spot

rate is £0.0151. Ninety-day interest rates available to Blades in the United Kingdom are 2%, while 90-day interest rates in Thailand are 3.75% (these rates are not annualized). Holt is aware that covered interest arbitrage, unlike locational and triangular arbitrage, requires an investment of funds. Thus, he would like to be able to estimate the British pound profit resulting from arbitrage over and above the amount available on a 90-day UK deposit.

Determine whether the forward rate is priced appropriately. If it is not priced appropriately, determine the profit you could generate for Blades

by withdrawing £100 000 from Blades' current account and engaging in covered interest arbitrage. Measure the profit as the excess amount above what you could generate by investing in the UK money market.

- 4 Why are arbitrage opportunities likely to disappear soon after they have been discovered? To illustrate your answer, assume that covered interest arbitrage involving the immediate purchase and forward sale of baht is possible. Discuss how the baht's spot and forward rates would adjust until covered interest arbitrage is no longer possible. What is the resulting equilibrium state called?

SMALL BUSINESS DILEMMA

Assessment of prevailing spot and forward rates by the Sports Exports Company

As the Sports Exports Company from Ireland exports basketballs to the United Kingdom, it receives British pounds. The cheque (denominated in pounds) for last month's exports just arrived. Jim Logan (owner of the Sports Exports Company) normally deposits the cheque with his local bank and requests that the bank convert the cheque to euros at the prevailing spot rate (assuming that he did not use a forward contract to hedge this payment). Jim's local bank provides foreign exchange services for many of its business customers who need to buy or sell widely traded currencies. Today, however, Jim decided to check the quotations of the spot rate at other banks before converting the payment into euros.

- 1 Do you think Jim will be able to find a bank that provides him with a more favourable spot rate than his local bank? Explain.
- 2 Do you think that Jim's bank is likely to provide more reasonable quotations for the spot rate of the British pound if it is the only bank in town that provides foreign exchange services? Explain.
- 3 Jim is considering using a forward contract to hedge the anticipated receivables in pounds next month. His local bank quoted him a spot rate of 1.45 euros and a one-month forward rate of 1.4435 euros. Before Jim decides to sell pounds one month forward, he wants to be sure that the forward rate is reasonable, given the prevailing spot rate. A one-month Treasury security in Ireland currently offers a yield (not annualized) of 1%, while a one-month Treasury security in the United Kingdom offers a yield of 1.4%. Do you believe that the one-month forward rate is reasonable given the spot rate of 1.45 euros?